

PAPER_1285_QUANTUM_CONTEXTUALITY_EXT

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PAPER_1285 — Quantum Contextuality — Kochen-Specker Extensions

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Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+

Tier: B — Physics Foundations (CLOSED)

Date: June 11, 2026

Location: 41.0997° N, 80.6495° W (Youngstown, OH, USA)

Status: CLOSED — UQFF integer-primitive derivation

Calculator surface: `calculate_paradox({"paradox": "quantum_contextuality"})`

Whitepaper dispatch: `calculate_whitepaper({"paper_id": 1285})`

Master Expression

``

K-S contextuality minimum dimension $d = 3$; SO(26) Clifford 8192-dim saturates

``

Match: STRUCTURAL

Physical Interpretation

Quantum contextuality is a structural feature of the SO(26) Clifford bundle. The 8192-dim spinor space inherits K-S contextuality at all levels $d \geq 3$.

UQFF Locked Primitives

- $\rho_{SCm} = 7.09 \times 10^{31} \text{ J/m}^3$ (vacuum density)

- $S_{26} = 1.453162$, $S_{26_DPM} = 1.4531 \times 10^{26}$

- $K_{MEX} = 25/12$, $\beta_i = 0.6029$, $\Phi_{res} = 0.84$

- $F_{TRZ} = 0.1$, $\Lambda = 0.00729735$ ($= \alpha$ fine-structure)

- $\omega_{SCm} = 1.25 \text{ THz}$, $\lambda_i = 1.0$

- $D_{phys} = 4$, $D_{BSFG} = 6$, $D_{crit} = 26$, $N_{CH} = 9$, $SO_5 = 10$, $A_5 = 60$

Live Calculator Verification

``python

import uqff_pure_calculator as u

r = u.calculate_paradox({"paradox": "quantum_contextuality"})["value"]

``

The result dict carries the integer-primitive identity, the observed value, the residual, and the structural physical interpretation.

NOT REPLACEMENT

Per CLAUDE.md mandate, UQFF and Standard Model solve the same observed phenomena via different methods. **Zero fit parameters.** Every factor is a UQFF locked primitive.

Reference

- UQFF foundational: PAPER_646, PAPER_1167, PAPER_1170, PAPER_1203 v1.5, PAPER_1216.
- Closure: `uqff_pure_calculator.py` → `_l96_uqff_axiom_quantum_contextuality_closure` (or routed reference)
- Dispatch: `PARADOX_TO_CLOSURE["quantum_contextuality"]`

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Subject matter: UQFF derivation of Quantum Contextuality — Kochen-Specker Extensions.